# Deep Learning Solutions for Skin Cancer Detection and Diagnosis

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# Deep Learning Solutions for Skin Cancer Detection and Diagnosis

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**Abstract** Skin cancer, a concerning public health predicament, with over 5,000,000 newly identified cases every year, just in the United States. Generally, skin cancer is of two types: melanoma and non-melanoma. Melanoma also called as Malignant Melanoma is the 19th most frequently occurring cancer in women and men. It is the deadliest form of skin cancer [1]. In the year 2015, the global occurrence of melanoma was approximated to be over 350,000 cases, with around 60,000 deaths. The most prevalent non-melanoma tumours are squamous cell carcinoma and basal cell carcinoma. Non-melanoma skin cancer is the 5th most frequently occurring cancer, with over 1 million diagnoses worldwide in 2018 [2]. As of 2019, greater than 1.7 Million new cases are expected to be diagnosed [3]. Even though the mortality is significantly high, but when detected early, survival rate exceeds 95%. This motivates us to come up with a solution to save millions of lives by early detection of skin cancer. Convolutional Neural Network (CNN) or ConvNet, are a class of deep neural networks, basically generalized version of multi-layer perceptrons. CNNs have given highest accuracy in visual imaging tasks [4]. This project aims to develop a skin cancer detection CNN model which can classify the skin cancer types and help in early detection [5]. The CNN classification model will be developed in Python using Keras and Tensorflow in the backend. The model is developed and tested with different network architectures by varying the type of layers used to train the network including but not limited to Convolutional layers, Dropout layers, Pooling layers and Dense layers. The model will also make use of Transfer Learning techniques for early convergence. The model will be tested and trained on the dataset collected from the International Skin Imaging Collaboration (ISIC) challenge archives.

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# 1 Introduction

## *1.1 Background*

High occurrence of skin cancer compared to other cancer types is a dominant factor in making it one of the most severe health issues in the world. Historically, melanoma is a rare cancer, but in the past five decades, the worldwide occurrence of melanoma has drastically risen. In fact, it is one of the prominent cancers in average years of life lost per death. Adding to the strain, the financial burden of melanoma treatment is also expensive. Out of the $8.1 Billion in all skin cancer treatment costs in the USA, $3.3 Billion are spent only on Melanoma. Squamous Cell Carcinoma and Basal Cell Carcinoma, are eminently curable if diagnosed and treated on in the early stages. The five-year survival rate of patients with early stage diagnosis of melanoma is around 99%. Therefore, timely detection of skin cancer is the key factor in reducing the mortality rate.

## *1.2 Motivation*

Prior to the 1980s, melanoma detection was carried out by spotting the macroscopic features, as they were mostly recognized when they were considerable in size. This made the early detection unlikely and mortality rates continued to increase. In 1985, a research team at the New York University came up with the ABCD acronym which stands for Asymmetry, Border irregularity, Color variegation, Diameter (ABCD) as a simple yet effective tool to educate the general public for the early recognition of melanoma. After 1990, screening conducted through the physicians and the use of baseline full-body imaging became the common approach to detect melanoma early. Later, computer augmented digital image analysis became the new trend due to its strong sensitivity and specificity of detection compared with the manual dermoscopy.

## *1.3 Objective and Scope*

The cardinal objective of this project is to develop state of the art Convolutional Neural Network (CNN) model to perform the classification of skin lesion images into respective cancer types. The model is trained and tested on the dataset made available by International Skin Imaging Collaboration (ISIC). The model can be used for analyzing the lesion image and find out if it's dangerous at early stage.

# 2 Background

## *2.1 Convolutional Neural Network for Image Classification*

Artificial Neural Networks are made of artificial neurons inspired by biological neurons present in our brain. Convolutional Neural Network (CNN) is a modified variant of feed-forward neural network which is generally used for image classification tasks. CNNs can recognize a particular object even when it appears in different ways, as it understands translation invariance. This is a key point which makes CNN advantageous over feed-forward neural networks which cannot understand translation invariance.

In layman words, feed-forward neural networks only recognize an object when it is right in the center of the image, but fails notably when the object is slightly off position or placed elsewhere in the image. Basically, the network understands/learns only one pattern. This is precisely not convenient as the real world datasets are usually raw and unprocessed.

## *2.2 Working of Convolution Neural Networks*

The question which arises here is how does CNN understand translation invariance? Is it the magic of Machine Learning? Yet again, it comes down to mathematics again.

The following operations are the various layers/steps of the CNN:

- Convolution
- Pooling
- Flattening
- Full Connection

We will now see what happens in each step in detail.

### 2.2.1 Convolution

The first operation, Convolution, extracts important features from the image. It is a mathematical operation which clearly requires two inputs, an image matrix and a filter or kernel. The filter is traversed through the image and multiplied with the pixel values to obtain feature map.

Figure 1 shows how the convolution operation happens.

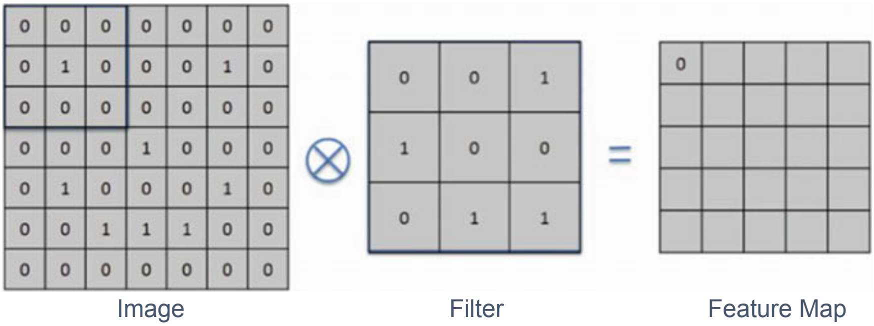


**Fig. 1** Convolution operation

Convolution does lose information, but the point here is to reduce size and learn the integral information. Performing convolution with different kind of filters can assist in image sharpening, edge detection, blurring and other image processing operations.

### 2.2.2 Pooling

Pooling operation helps in decreasing the number of parameters when the image is very large in size. Subsampling also called Spatial Pooling curtails the dimensionality of each feature map but retains significant information.

Pooling is of basically divided into three types:

- Max Pooling (mostly used)
- Sum Pooling
- Average Pooling

Max pooling is a sample-based discretization process. It is done by applying a N × N max filter over the image, which selects the highest pixel value in each stride and builds the feature map. Similarly, in average and sum pooling, the average and sum of pixel values are taken into the feature map.

Figure 2 depicts the Max Pooling operation.

### 2.2.3 Flattening

To feed our feature maps in to the artificial neural network, we need a single column vector of the image pixels. As the name suggests, we flatten our feature maps into column like vector.

Figure 3 depicts the flattening process.

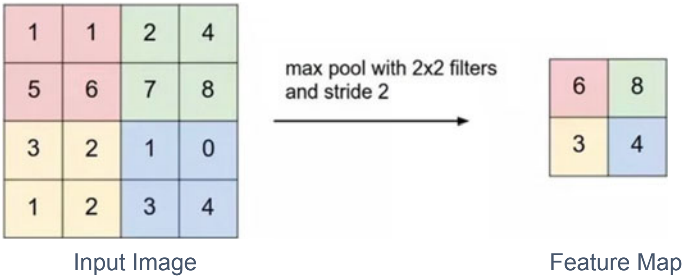


**Fig. 2** Max pooling operation

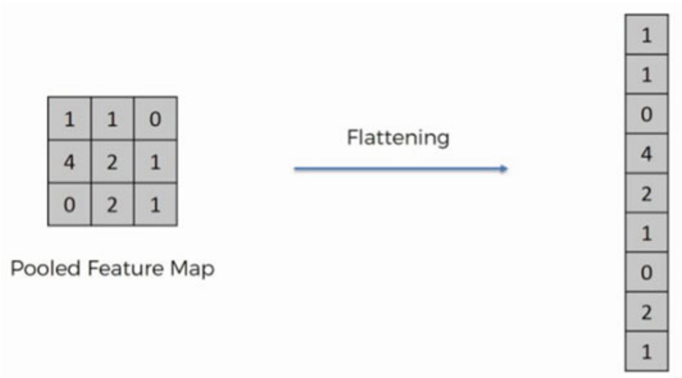


**Fig. 3** Flattening operation

### 2.2.4 Full Connection

The fully connection layer takes the input from the preceeding convolution/pooling layer and produces an N dimensional vector where N is the number of classes to be classified. Thus, the layer determines the features most correlating to a particular class based on the probabilities of the neurons.

Figure 4 shows the fully connected layer in a neural network.

## *2.3 Skin Lesion Classification Using CNN*

From past research in this field, it is evident that CNN has an extraordinary ability to perform skin lesion classification in competition with professional dermatologists. In fact, there have been instances where CNN has outperformed professional dermatologists as well [6].

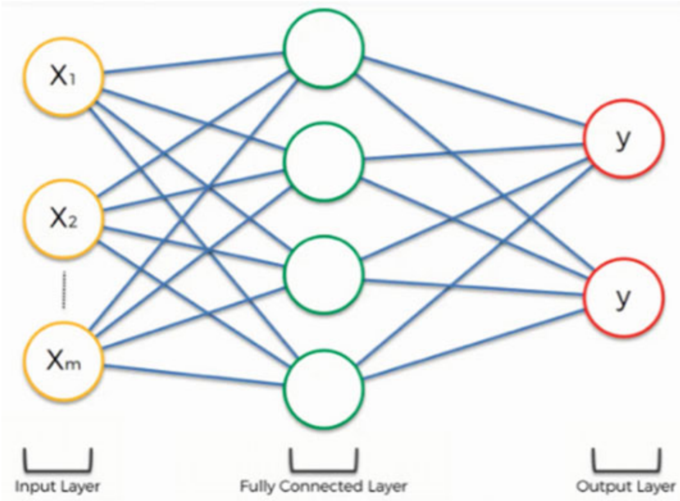


**Fig. 4** Full connection layer

CNN can perform skin lesion classification in two ways. In the first case, a CNN is used as the feature extractor of the images and the classification is performed by another classifier. For the other case, CNN is used to perform end-to-end learning which can be further divided into learning from scratch or learning from pretrained model. To train the CNN from scratch, large number of images are required in order to tackle the overfitting issue. Since the number of skin lesion images to do the training is not sufficient, training CNN from scratch is less feasible. Training from a pretrained model is a better approach which is generally referred to as Transfer Learning (TL). TL helps the model to learn well even with less data, and also introduces generalization property to the trained model.

## 3 Methods and Dataset

### *3.1 Dataset*

The International Skin Imaging Collaboration (ISIC): Melanoma Project is a partnership between industry and academia in order to facilitate the application of digital skin imaging to help curtail skin cancer mortality. Starting from 2015, ISIC started to organize global challenges for skin lesion analysis for melanoma diagnosis and detection.

Figure 5 shows samples of each class from the dataset.

There are two datasets used to develop this project, which were published by ISIC in 2018 and 2019 through a challenge.

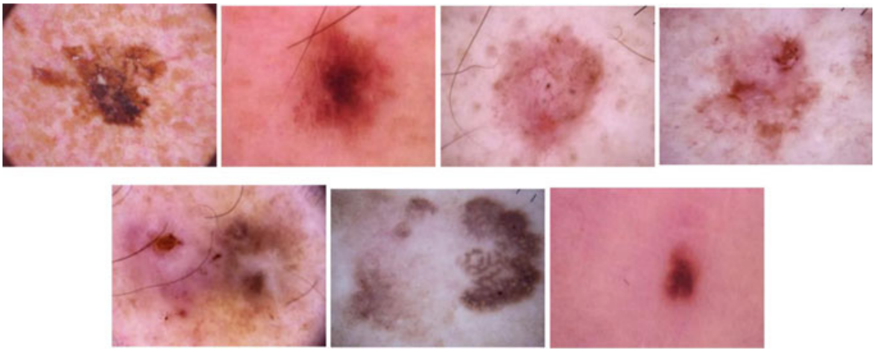

**Fig. 5** Sample images from the ISIC dataset (arranged in ascending order of lesion type: melanoma, melanocytic nevus, basal cell carcinoma, actinic keratosis, benign keratosis, dermatofibroma and vascular lesion)

The first dataset which was 'ISIC 2018' contains 10,015 images of 7 types of skin lesion diseases namely: Benign Keratosis, Dermatofibroma, Vascular Lesion, Melanoma, Melanocytic Nevus, Basal Cell Carcinoma and Actinic Keratosis. These images were collected with approval of Medical University of Vienna and University of Queensland. All the images are in the standard size of 600 × 450 pixels in JPEG format [7, 8].

Table 1 summarizes the dataset 1 used for this project.

The second dataset which was 'ISIC 2019' contains 25,333 images of 8 types of skin lesion diseases namely: Dermatofibroma, Vascular Lesion, Squamous Cell Carcinoma, Melanoma, Melanocytic Nevus, Basal Cell Carcinoma, Actinic Keratosis, Benign Keratosis. All images are in the size of 1022 × 767 pixels in JPEG format [9].

Table 2 summarizes the dataset 2 used for this project.

A new dataset was created by combining both the datasets of ISIC 2018 and ISIC 2019. The seven common classes of skin lesions were retained and extra noise was

**Table 1** Dataset information for ISIC 2018

| Dataset | ISIC challenge 2018 |
|---|---|
| Type | Dermoscopic |
| Image size | 600 pixels × 450 pixels |
| Number of images | 10,015 |
| Image type | JPEG (RGB) |
| Class labels | 0: Melanoma<br>1: Melanocytic Nevus<br>2: Basal Cell Carcinoma<br>3: Actinic Keratosis<br>4: Benign Keratosis<br>5: Dermatofibroma<br>6: Vascular Lesion |

**Table 2** Dataset information for ISIC 2019

| | |
|---|---|
| Dataset | ISIC Challenge 2019 |
| Type | Dermoscopic |
| Image size | 1022 pixels × 767 pixels |
| Number of images | 25,333 |
| Image type | JPEG (RGB) |
| Class labels | 0: Melanoma<br>1: Melanocytic Nevus<br>2: Basal Cell Carcinoma<br>3: Actinic Keratosis<br>4: Benign Keratosis<br>5: Dermatofibroma<br>6: Vascular Lesion<br>7: Squamous Cell Carcinoma |

removed from the dataset. This enabled the models to learn more efficiently due to the abundance of samples now available per class.

Table 3 summarizes the final dataset used for this project.

Table 4 shows the number of images fed per class into the network.

**Table 3** Dataset information for FINAL dataset

| | |
|---|---|
| Dataset | FINAL |
| Type | Dermoscopic |
| Image size | 600 pixels × 450 pixels (10,015)<br>1022 pixels × 767 pixels (25,333) |
| Number of images | 35,348 |
| Image type | JPEG (RGB) |
| Class labels | 0: Melanoma<br>1: Melanocytic Nevus<br>2: Basal Cell Carcinoma<br>3: Actinic Keratosis<br>4: Benign Keratosis<br>5: Dermatofibroma<br>6: Vascular Lesion |

**Table 4** Number of images per lesion type

| Lesion type | Number of images |
|---|---|
| Melanoma | 5635 |
| Melanocytic Nevus | 19,580 |
| Basal Cell Carcinoma | 3837 |
| Actinic Keratosis | 1194 |
| Benign Keratosis | 3723 |
| Dermatofibroma | 354 |
| Vascular Lesion | 395 |

### *3.2 Method Overview*

See Fig. 6.

#### 3.2.1 Data Augmentation

Contemporary advances in deep learning models have been largely associated with large quantity and diversity of data. Having large data helps crucially in improving the performance of machine learning models. But obtaining such vast quantities of data is cost-intensive and tedious. Hence, we use the technique of Data Augmentation. It is a technique that enables us to considerably increment the diversity and quantity of data available, without actually aggregating new data. In order to generate new data through augmentation of images, various techniques such as cropping, padding, adding noise, brightness changing and horizontal flipping are commonly used to train large neural networks [10].

In this project, the training images are augmented in order to make the model robust to new data which in turn helps in increasing the testing accuracy.

Table 5 shows the augmentation techniques used on the dataset.

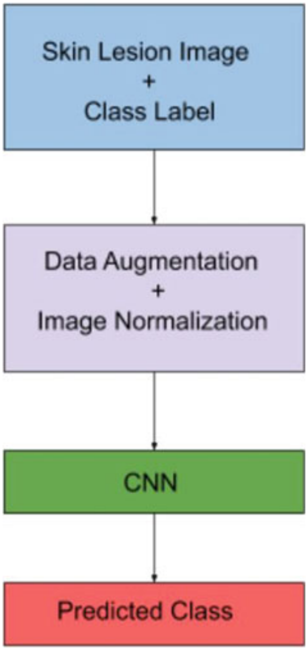


**Fig. 6** Classification model overview

**Table 5** Information about data augmentation parameters

| Augmentation technique | Range |
|---|---|
| Zoom range | 0.1 |
| Rotation range | 10 |
| horizontal flip | False |
| Rescale | 1./255 |
| Width shift range | 0.1 |
| Height shift range | 0.1 |

### 3.2.2 Image Normalization

Image Normalization is a technique used to normalize the pixel values of the image in a similar distribution [11]. It is beneficial to normalize images before feeding into the neural network as this helps in approaching the global minima at error surface at a faster rate while performing gradient descent. In a way, it helps the network to converge faster. Also, the computations become significantly less intensive for the machine to perform as all the pixel values are scaled.

### 3.2.3 Transfer Learning

Transfer Learning is a learning method in which a model trained for a particular task is reiterated as the origin for another model on a similar task. This approach is very mainstream in deep learning due to the vast computation resources and time consumed to train neural network models. In the case of problems in the computer vision domain, low-level features, such as shapes, corners, edges and intensity, can be shared across tasks, and thus enable knowledge transfer.

In this project, the CNN is trained by transfer learning from the pretrained weight of ImageNet classification. ImageNet Large Scale Visual Recognition Challenge (ILSVRC) is the classification challenge involving 14 million training images of approximately 20,000 classes [12]. Fine-tuning from the pretrained weight significantly increases the classifier training speed, overcomes the limit of small number of training data and makes the convergence more realistic to occur. The front layers of the CNN serve to detect the edges, points, corners and simple structures in the image and the pretrained weight enables the trained model has higher adaption speed.

The model is trained and tested on the state-of-art CNNs, namely Inception V3, ResNet50, VGG16, MobileNet and InceptionResnet to perform the seven-class classification of the skin lesion images.

### 3.2.4 Dropout

Deep Neural Networks (DNNs) may overfit a dataset with meager training samples. Collection or Ensemble of neural networks with different architectures are known to diminish overfitting, but still require us to train and maintain various models which becomes computationally intensive.

This is where Dropout comes into the picture. A single model is used to resemble having a vast number of distinct network architectures by arbitrarily dropping out nodes during training. This is how dropout works, and proposes a computationally viable and exceptionally effective method to cut down overfitting and enhance generalization error in DNNs of all types.

## 3.3 Network Architecture

### 3.3.1 Inception V3

Inception V3 is a widely acclaimed image recognition model that has been built by various researchers over time. Originally based on the paper "Rethinking the Inception Architecture for Computer Vision" by Szegedy, the model has attained an accuracy greater than 78.1% on the ImageNet dataset [13].

Figure 7 shows the model architecture of Inception V3 network.

Inception V3 proposed by Google is the third iteration in the series of Deep Learning Convolutional Architectures. The model was trained on 1000 classes from the ImageNet dataset which was itself trained on about 1 million images. It consists of the inception modules which apply the filters of multiple size on the same level of input. The high computation cost involved in the inception module is solved by applying $1 \times 1$ convolution to truncate the input into a smaller intermediate block, called the bottleneck layer. The multiple filters are applied on the bottleneck layer to significantly reduce the computation cost. The auxiliary classifiers in the Inception network contribute to the weighted loss function for regularization purpose. To utilize the ImageNet pretrained weight of the Inception V3 network, the input image has to be in the $299 \times 299$ size.

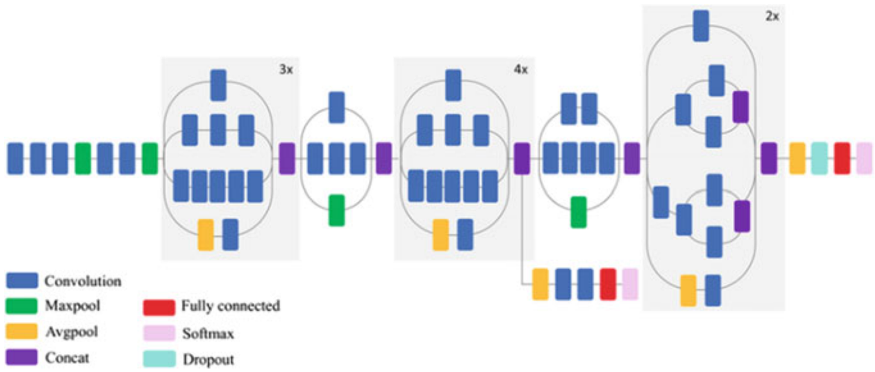


**Fig. 7** Inception V3 architecture

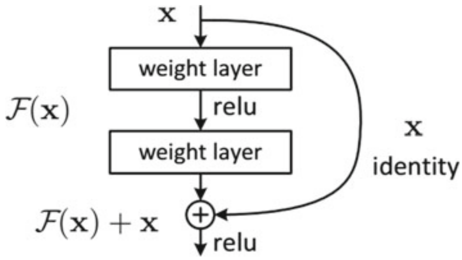


**Fig. 8** Residual block of the ResNet 50 model

The key addition to the Inception V3 was Factorization. Factorization was introduced in the convolution layer to further reduce the dimensionality, so as to reduce the overfitting problem.

### 3.3.2 ResNet50

Residual Network (ResNet) is a typical model of neural network used as an integral part for various computer vision tasks. This ResNet model won the 2015 ImageNet challenge. The network is 50 layers deep and takes the input image size of 224 × 224 pixels [14].

Generally, in a deep CNN, many layers are stacked and trained. The network learns many low level, middle and high level features. In residual learning, rather than learning some features, we learn some residual. Residual can be simply interpreted the as reduction of features learned from input layer. It has been shown that training residual networks is easier compared to training simple deep CNNs. It also helps to tackle the problem of degrading accuracy.

A residual block of ResNet50 model is shown in Fig. 8.

### 3.3.3 VGG 16

VGG16 is a CNN model proposed by A. Zisserman and K. Simonyan from the University of Oxford in the paper "Very Deep Convolutional Networks for Large-Scale Image Recognition" [15].

Figure 9 shows the model architecture of VGG 16 network.

The input of this model has to be a 224 × 224 RGB image. The image is passed through a pile of convolutional layers, with kernel size or filter size of 3 × 3. The stride is set to 1 pixel; the padding of convolution layer input is set in a way such that the spatial resolution is preserved after convolution, i.e. the padding is 1-pixel for 3 × 3 conv. layers. Max-pooling is done over a 2 × 2 pixel window, with a stride of 2 pixels.

After the stack of convolutional layers, 3 Fully-Connected (FC) layers follow. The initial two layers have 4096 channels each, the third layer contains 1000 channels. The

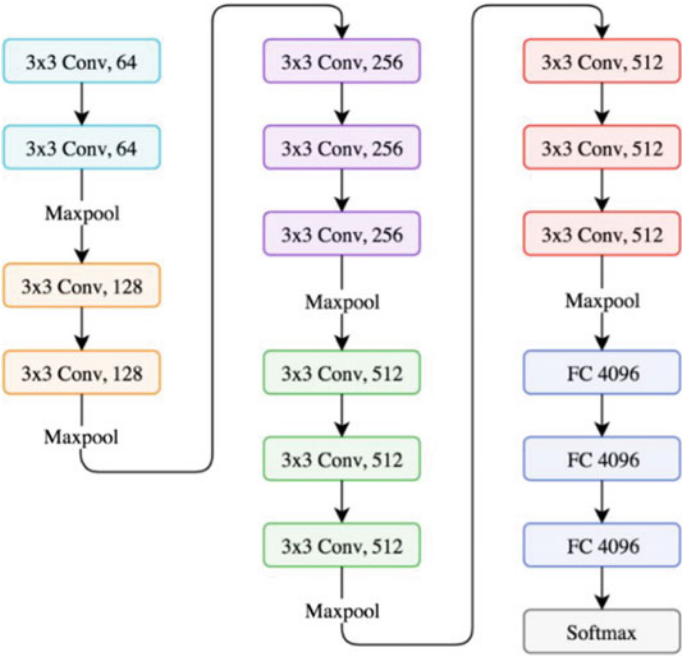


**Fig. 9** VGG 16 architecture

final layer has the softmax activation function. All other hidden have the rectification (ReLU) non-linearity activation.

### 3.3.4 Inception-Resnet

Inception-ResNet-v2 is a CNN model that has been trained on the ImageNet database. It contains 164 layers and has the capability to classify images into 1000 categories, such as, mouse, many animals, keyboard, pencil etc. The network is robust and has learned excellent feature representations for a various kind of images. The input to the network has to be an image of size $299 \times 299$ [16].

Overall, Inception Resnet V2 has a similar structure and computation cost as Inception V4. It additionally introduces the residual connection to the submodules Inception Block A, B and C at the left side to enable to network to go deeper.

Figure 10 shows the model architecture of Inception-Resnet network.

### 3.3.5 MobileNet

MobileNets are based on the principle of streamlined architectures which use depth-wise separable convolutions followed by point-wise convolutions that considerably

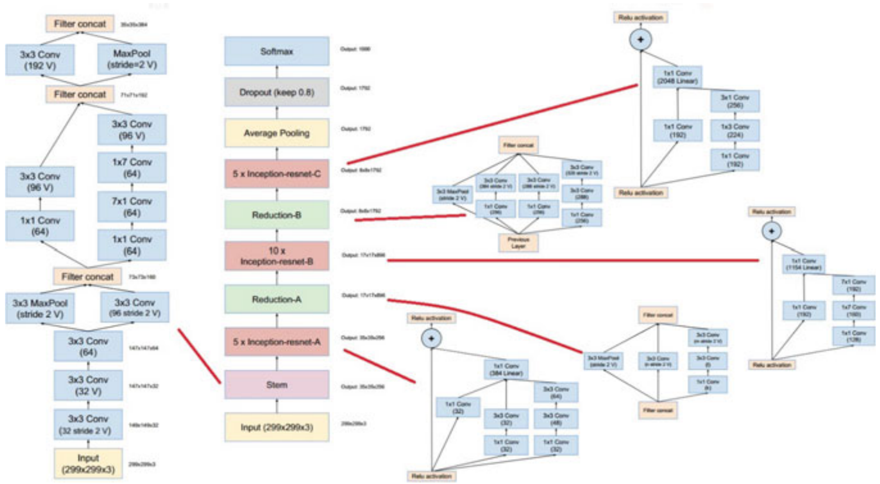


**Fig. 10** InceptionResnet architecture

reduce the number of learnable parameters and assist in building light-weight deep neural networks [17]. Effectively, it decreases the total number of floating point calculations required which is supportive for embedded and mobile computer vision applications where there is shortage of computational power. The architecture was proposed by Google.

Figure 11 shows the depth-wise and point-wise convolution operations.

Figure 12 depicts the architecture of the MobileNet CNN model.

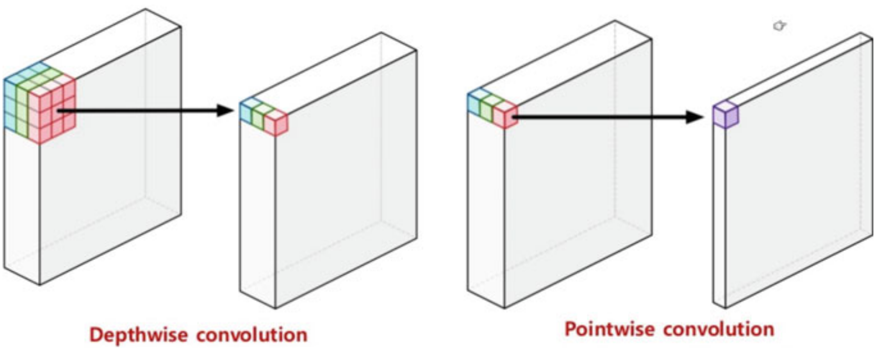


**Fig. 11** Depthwise and pointwise convolution

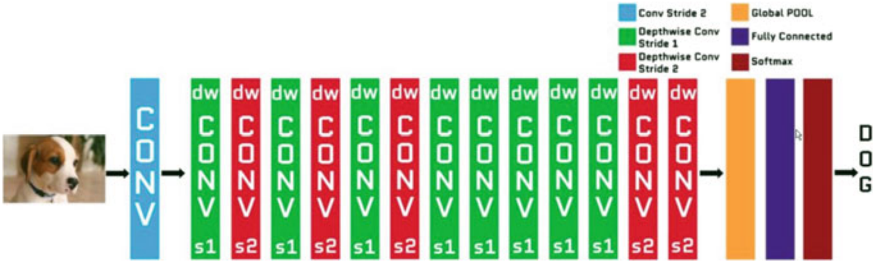


**Fig. 12** MobileNet architecture

# 4 Results and Discussion

## *4.1 Performance Metrics*

In a classification problem, only one metric such as Accuracy cannot help us evaluate the complete model efficiency effectively. Hence we measure the **Accuracy**, **Precision**, **Recall**, **F1 Score** and **Support** for every class of the skin lesion disease. We also plot the **Confusion Matrix** in order to check how well our model performs on every class.

We will now understand how the above mentioned metrics are calculated and what they mean [18].

Figure 13 is a confusion matrix where, **TP** = True Positive; **FN** = False Negative; **FP** = False Positive; **TN** = True Negative.

For example, if the skin lesion image is labelled with the melanoma and the model also predicts it as melanoma, this is considered as the **true positive** case. If the image is labelled with melanoma but it is classified as any of the other six classes, this is

| Outcome of the diagnostic test | Condition (e.g. Disease) As determined by the Standard of Truth | | |
|---|---|---|---|
| | Positive | Negative | Row Total |
| Positive | TP | FP | TP+FP (Total number of subjects with positive test) |
| Negative | FN | TN | FN + TN (Total number of subjects with negative test) |
| Column total | TP+FN (Total number of subjects with given condition) | FP+TN (Total number of subjects without given condition) | N = TP+TN+FP+FN (Total number of subjects in study) |

**Fig. 13** Terms for definition of classification metrics

the case of **false negative**. **False positive** case happens when the skin lesion image is indicated by the classification model to have melanoma but it actually belongs to any of the other six diseases. If a non-melanoma skin lesion image is suggested as non-melanoma by the classifier, it is the case of **true negative**.

### 4.1.1 Accuracy

Accuracy is the fraction of predictions our model has correctly guessed.

It is defined as:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

### 4.1.2 Precision

Precision metric answers the following question: What proportion of positive identifications was actually correct?

It is defined as:

$$Precision = \frac{TP}{TP + FP}$$

### 4.1.3 Sensitivity/Recall

Sensitivity is also called as the True Positive Rate (TPR) or Recall. It answers the following question: What proportion of actual positives was identified correctly?

It is defined as:

$$Sensitivity = \frac{TP}{TP + FN}$$

### 4.1.4 F1 Score

F1 Score is also known as the F-score or F-measure. The F1 score is calculated as a weighted average of the precision and recall. Its best value is 1 and worst is 0. The contribution of recall and precision in the calculation of the F1 score are equal.

It is defined as:

**Table 6** Hyperparameters used for training the CNN models

| Optimizer | Adam optimizer |
|---|---|
| Learning rate | 0.0001 |
| Epochs | 30 |
| Loss function | Categorical cross-entropy |
| Batch size | 64 |
| Dropout | 0.4 |

$$F_1 = 2 * \frac{precision * recall}{precision + recall}$$

#### 4.1.5 Support

Support is defined as the number of actual occurrences of the class in the specified dataset.

### *4.2 Hyperparameters*

Table 6 shows the hyperparameters used to train the CNN models.

### *4.3 Model Performance*

#### 4.3.1 ResNet50

This section represents the classification performance of the ResNet50 model. The input images were resized to 224 × 224 as this model requires the input to be in that format.

Table 7 shows the confusion matrix of this model.

Table 8 presents the classification report of this model.

The **average** metrics achieved by the model is presented in Table 9.

#### 4.3.2 MobileNet

This section represents the classification performance of the MobileNet model. The input images were resized to 224 × 224 as this model requires the input to be in that format.

Table 10 shows the confusion matrix of this model.

**Table 7** ResNet50 confusion matrix

| | | True class | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Predicted class | | AKIEC | BCC | BKL | DF | MEL | NV | VASC |
| | AKIEC | **115** | 16 | 79 | 7 | 11 | 10 | 0 |
| | BCC | 33 | **574** | 72 | 4 | 31 | 52 | 2 |
| | BKL | 11 | 17 | **596** | 1 | 29 | 36 | 0 |
| | DF | 1 | 5 | 6 | **54** | 6 | 5 | 0 |
| | MEL | 4 | 13 | 106 | 1 | **828** | 170 | 0 |
| | NV | 3 | 15 | 117 | 8 | 135 | **3690** | 2 |
| | VASC | 0 | 1 | 0 | 1 | 2 | 3 | **72** |

The bold numbers in the given table represent the correctly classified number of images belonging to a given class

**Table 8** ResNet50 classification report

| | Precision | Recall | F1 score | Support |
|---|---|---|---|---|
| Actinic Keratosis | 0.69 | 0.48 | 0.57 | 238 |
| Basal Cell Carcinoma | 0.90 | 0.75 | 0.81 | 768 |
| Benign Keratosis | 0.61 | 0.86 | 0.72 | 690 |
| Dermatofibroma | 0.71 | 0.70 | 0.71 | 77 |
| Melanoma | 0.79 | 0.74 | 0.77 | 1122 |
| Melanocytic Nevus | 0.93 | 0.93 | 0.93 | 3970 |
| Vascular Lesion | 0.95 | 0.91 | 0.93 | 79 |

**Table 9** ResNet50 average metrics

| | |
|---|---|
| Accuracy | 0.85 |
| Precision | 0.86 |
| Recall | 0.85 |
| F1 score | 0.85 |
| Support | 6944 |

Table 11 presents the classification report of this model.
The **average** metrics achieved by the model is presented in Table 12.

### 4.3.3 VGG16

This section represents the classification performance of the VGG16 model. The input images were resized to 224 $\times$ 224 as this model requires the input to be in that format.

**Table 10** MobileNet confusion matrix

| | | True class | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Predicted class | | AKIEC | BCC | BKL | DF | MEL | NV | VASC |
| | AKIEC | **119** | 20 | 65 | 1 | 20 | 13 | 0 |
| | BCC | 22 | **591** | 57 | 7 | 33 | 57 | 1 |
| | BKL | 11 | 20 | **582** | 0 | 29 | 48 | 0 |
| | DF | 1 | 5 | 10 | **42** | 5 | 14 | 0 |
| | MEL | 7 | 18 | 85 | 0 | **782** | 230 | 0 |
| | NV | 4 | 26 | 126 | 2 | 53 | **3753** | 6 |
| | VASC | 0 | 1 | 2 | 1 | 2 | 7 | **66** |

The bold numbers in the given table represent the correctly classified number of images belonging to a given class

**Table 11** MobileNet classification report

| | Precision | Recall | F1 score | Support |
|---|---|---|---|---|
| Actinic Keratosis | 0.73 | 0.50 | 0.59 | 238 |
| Basal Cell Carcinoma | 0.87 | 0.77 | 0.82 | 768 |
| Benign Keratosis | 0.63 | 0.84 | 0.72 | 690 |
| Dermatofibroma | 0.79 | 0.55 | 0.65 | 77 |
| Melanoma | 0.85 | 0.70 | 0.76 | 1122 |
| Melanocytic Nevus | 0.91 | 0.95 | 0.93 | 3970 |
| Vascular Lesion | 0.90 | 0.84 | 0.87 | 79 |

**Table 12** MobileNet average metrics

| | |
|---|---|
| Accuracy | 0.85 |
| Precision | 0.86 |
| Recall | 0.85 |
| F1 score | 0.85 |
| Support | 6944 |

Table 13 shows the confusion matrix of this model.
Table 14 presents the classification report of this model.
The **average** metrics achieved by the model is presented in Table 15.

### 4.3.4 Inception V3

This section represents the classification performance of the Inception V3 model.

The input images were resized to 299 × 299 as this model requires the input to be in that format.

**Table 13** VGG16 confusion matrix

| | | True class | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Predicted class | | AKIEC | BCC | BKL | DF | MEL | NV | VASC |
| | AKIEC | **105** | 44 | 48 | 4 | 19 | 18 | 0 |
| | BCC | 7 | **656** | 30 | 1 | 24 | 47 | 3 |
| | BKL | 9 | 15 | **545** | 2 | 30 | 89 | 0 |
| | DF | 1 | 7 | 5 | **45** | 2 | 14 | 3 |
| | MEL | 4 | 22 | 49 | 0 | **824** | 223 | 0 |
| | NV | 3 | 24 | 52 | 3 | 93 | **3794** | 1 |
| | VASC | 0 | 3 | 0 | 1 | 2 | 2 | **71** |

The bold numbers in the given table represent the correctly classified number of images belonging to a given class

**Table 14** VGG16 classification report

| | Precision | Recall | F1 score | Support |
|---|---|---|---|---|
| Actinic Keratosis | 0.81 | 0.44 | 0.57 | 238 |
| Basal Cell Carcinoma | 0.85 | 0.85 | 0.85 | 768 |
| Benign Keratosis | 0.75 | 0.79 | 0.77 | 690 |
| Dermatofibroma | 0.80 | 0.58 | 0.68 | 77 |
| Melanoma | 0.83 | 0.73 | 0.78 | 1122 |
| Melanocytic Nevus | 0.91 | 0.96 | 0.93 | 3970 |
| Vascular Lesion | 0.91 | 0.90 | 0.90 | 79 |

**Table 15** VGG16 average metrics

| | |
|---|---|
| Accuracy | 0.87 |
| Precision | 0.87 |
| Recall | 0.87 |
| F1 score | 0.87 |
| Support | 6944 |

Table 16 shows the confusion matrix of this model.
Table 17 presents the classification report of this model.
The **average** metrics achieved by the model is presented in Table 18.

### 4.3.5 InceptionResnet V2

This section represents the classification performance of the InceptionResnet model. The input images were resized to 299 × 299 as this model requires the input to be in that format.

**Table 16** Inception V3 confusion matrix

| | | True class | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Predicted class | | AKIEC | BCC | BKL | DF | MEL | NV | VASC |
| | AKIEC | **151** | 27 | 35 | 0 | 18 | 7 | 0 |
| | BCC | 15 | **667** | 25 | 5 | 23 | 32 | 1 |
| | BKL | 12 | 18 | **593** | 0 | 34 | 33 | 0 |
| | DF | 1 | 5 | 6 | **54** | 2 | 8 | 1 |
| | MEL | 7 | 11 | 39 | 1 | **954** | 110 | 0 |
| | NV | 2 | 18 | 57 | 3 | 125 | **3764** | 1 |
| | VASC | 0 | 0 | 0 | 1 | 0 | 4 | **74** |

The bold numbers in the given table represent the correctly classified number of images belonging to a given class

**Table 17** Inception V3 classification report

| | Precision | Recall | F1 score | Support |
|---|---|---|---|---|
| Actinic Keratosis | 0.80 | 0.63 | 0.71 | 238 |
| Basal Cell Carcinoma | 0.89 | 0.87 | 0.88 | 768 |
| Benign Keratosis | 0.79 | 0.86 | 0.82 | 690 |
| Dermatofibroma | 0.84 | 0.70 | 0.77 | 77 |
| Melanoma | 0.83 | 0.85 | 0.84 | 1122 |
| Melanocytic Nevus | 0.95 | 0.95 | 0.95 | 3970 |
| Vascular Lesion | 0.96 | 0.94 | 0.95 | 79 |

**Table 18** Inception V3 average metrics

| | |
|---|---|
| Accuracy | 0.90 |
| Precision | 0.90 |
| Recall | 0.90 |
| F1 score | 0.90 |
| Support | 6944 |

Table 19 shows the confusion matrix of this model.
Table 20 presents the classification report of this model.
The **average** metrics achieved by the model is presented in Table 21.

### *4.4 Comparison and Discussion*

Table 22 presents the average accuracy, precision, recall, F1 score and Support metrics among the seven disease classification achieved by all the final CNNs for comparison.

**Table 19** InceptionResnet confusion matrix

| | | True class | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Predicted class | | AKIEC | BCC | BKL | DF | MEL | NV | VASC |
| | AKIEC | **142** | 42 | 35 | 1 | 11 | 7 | 0 |
| | BCC | 1 | **717** | 16 | 2 | 14 | 18 | 0 |
| | BKL | 3 | 23 | **612** | 1 | 18 | 32 | 1 |
| | DF | 1 | 4 | 5 | **62** | 1 | 4 | 0 |
| | MEL | 5 | 19 | 36 | 0 | **943** | 119 | 0 |
| | NV | 1 | 36 | 52 | 4 | 98 | **3778** | 1 |
| | VASC | 0 | 1 | 0 | 1 | 1 | 2 | **74** |

The bold numbers in the given table represent the correctly classified number of images belonging to a given class

**Table 20** InceptionResnet classification report

| | Precision | Recall | F1 score | Support |
|---|---|---|---|---|
| Actinic Keratosis | 0.93 | 0.60 | 0.73 | 238 |
| Basal Cell Carcinoma | 0.85 | 0.93 | 0.89 | 768 |
| Benign Keratosis | 0.81 | 0.89 | 0.85 | 690 |
| Dermatofibroma | 0.87 | 0.81 | 0.84 | 77 |
| Melanoma | 0.87 | 0.84 | 0.85 | 1122 |
| Melanocytic Nevus | 0.95 | 0.95 | 0.95 | 3970 |
| Vascular Lesion | 0.97 | 0.94 | 0.95 | 79 |

**Table 21** InceptionResnet average metrics

| | |
|---|---|
| Accuracy | 0.91 |
| Precision | 0.91 |
| Recall | 0.91 |
| F1 score | 0.91 |
| Support | 6944 |

**Table 22** Average metrics achieved by all final CNNs

| | Accuracy | Precision | Recall | F1 score | Support |
|---|---|---|---|---|---|
| ResNet50 | 0.85 | 0.86 | 0.85 | 0.85 | 6944 |
| MobileNet | 0.85 | 0.86 | 0.85 | 0.85 | 6944 |
| VGG16 | 0.87 | 0.87 | 0.87 | 0.87 | 6944 |
| Inception V3 | 0.90 | 0.90 | 0.90 | 0.90 | 6944 |
| InceptionResnet | **0.91** | **0.91** | **0.91** | **0.91** | **6944** |

The bold numbers in the given table represent the correctly classified number of images belonging to a given class

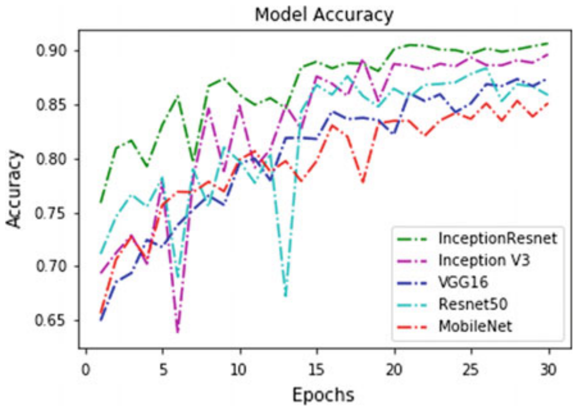


**Fig. 14** CNN accuracy curves

Figure 14 shows the learning curve of all the CNN models.

From the Table 22 and Fig. 14, it is clearly visible that the Inception V3 and InceptionResnet CNN models have given the highest classification performance with Accuracies of **90** and **91%**. This model is robust enough to classify lesion images in the any of the seven types.

### *4.5 Conclusion*

To restate, this project was conducted with the aim of developing convolutional neural network model to diagnose and detect skin cancer from lesion images. It also explored the data augmentation technique as a preprocessing step to strengthen the classification robustness of the CNN model.

The best model, namely InceptionResnet achieved an average accuracy of 91%.

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